

Agent Guardrails — Anti-Forgetting Enforcement (§11.4.109)

Revision: 1 **Last modified:** 2026-07-09T00:00:00Z **Classification:** consumer instantiation of the universal §11.4.109 mandate. The canonical preamble + checklist pattern lives in `constitution/docs/AGENT_GUARDRAILS.md`; this file is the Helix OTA project-level instantiation (§11.4.35) and maps the guardrails to *this* project's concrete surfaces.

Mandatory constraints MUST NOT depend on an agent remembering them. Two layers make the rules mechanical (§11.4.109):

1. **`constitution/scripts/hooks/guard-forbidden-commands.sh`** — a Claude Code PreToolUse hook (inherited by reference, NEVER copied — a copy diverges silently). Wired in `.claude/settings.json`. It BLOCKS at the tool-call boundary, no matter what any agent remembers: `git push --force|-f|--force-with-lease|--no-verify|--no-gpg-sign` (§11.4.113 absolute no-force-push), `sudo/su` (§11.4.161 rootless), raw host-direct emulator/adb, and host power-management (§12 Host-Session Safety). Escape hatch: a command carrying the literal `# guardrails:allow <reason>` marker is warned but allowed — EXCEPT host-power, which is never overridable.
2. **This document** — the preamble the orchestrator pastes into every subagent dispatch, and the checklist the orchestrator runs before every state-changing action.

SUBAGENT CONSTITUTIONAL PREAMBLE

Paste this verbatim into every subagent dispatch (§11.4.20/§11.4.70 subagent-driven default):

You operate under the Helix Constitution (`constitution/`). Non-negotiable, mechanically enforced (§11.4.109): - **No bluff (§11.4)**. Every claimed result carries real captured physical evidence

(§11.4.5/§11.4.69/§11.4.107). Metadata-only / config-only / absence-of-error / grep-without- runtime PASS are forbidden. If you cannot prove it, say UNKNOWN: / PENDING_FORENSICS:. - **No force-push, ever (§11.4.113)**. No history rewrite, no --no-verify. Integrate onto latest main and fast-forward. - **Rootless only (§11.4.161)**. No sudo/su. Containers via the vasic-digital/containers submodule (§11.4.76), Podman rootless. - **Target + host safety (§11.4.133 / §12)**. Never brick the target device or destabilize the host (no suspend/poweroff/logout; ≤60% host memory §12.6). - **Stay in scope**. Edit only the files named in your task. Do NOT run git add/commit/push unless the orchestrator explicitly delegates it — the orchestrator serializes commits (§11.4.84 working-tree quiescence). - **Systematic debugging first (§11.4.102)**. No fix without a proven root cause.

ORCHESTRATOR PRE-ACTION CHECKLIST

Run before every state-changing action (commit, push, tag, build, flash, deploy):

1. **Fetch-before-edit (§11.4.37)**: git fetch --all --prune on parent + owned submodules ran this session; integrate any incoming before editing.
 2. **Quiescence (§11.4.84)**: before git add, grep the working tree for mutation residue (MUTATED for paired, // always pass, _mutated_); every staged file accounted for vs the declared scope; no in-flight paired-mutation gate live.
 3. **Every change reviewed (§11.4.142)**: an independent reviewer (subagent, not the author) has cleared the diff to a zero-finding GO (§11.4.134) before commit/build.
 4. **Validate before commit (§11.4.26 step 3 / §11.4.40)**: inheritance gate + pre-build sweep + regression + meta GREEN on the real tree.
 5. **No force-push (§11.4.113)**: pushes are fast-forward onto latest main; fan out to all four upstreams (§2.1).
 6. **Multi-agent utilization (§11.4.183 / §11.4.103)**: keep ≥3 non-contending background streams busy while the main stream is free; idle only when genuinely blocked (§11.4.94/§11.4.101).
 7. **Manual-QA final confirmation (§11.4.185)**: a release tag additionally requires the manual QA-team confirmation-of-done step.
-

Helix OTA project-specific surfaces the guardrails protect

- **Containers (§11.4.76/§11.4.161/§11.4.173):** all container + build workloads go through submodules/containers (rootless Podman); no bare-host or sudo-escalated container runs.
- **Target-hardware safety (§11.4.133):** RK3588 / Orange Pi 5 Max flashing + A/B updates use the sanctioned tools + integrity-verified images only; no unverified voltage/clock/regulator writes.
- **Trust boundary (project CLAUDE.md):** the artifact-signature verification key comes ONLY from server config, never from the request.

Cross-reference: constitution/docs/AGENT_GUARDRAILS.md (canonical), .claude/settings.json (hook wiring), constitution/scripts/hooks/guard-forbidden-commands.sh (the mechanism).